

Course : MME 302

Exp. No : 05

Exp. Name : Cast Iron foundry practice

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objective :-

To cast irons using different raw materials and additives and to study the effects of process variables and ~~addit~~ on the nature and properties of cast iron.

Theory:

Cast iron is an alloy of C and Fe with C content above 2%. Its excellent castability and mechanical properties make it suitable for a wide range of applications. The main types of cast iron are gray, white, ductile and malleable. Types of cast iron differ based on their carbon form and microstructure. But, microstructure is influenced by raw material composition and cooling conditions.

During solidification, carbon may either form graphite or combine with iron to produce cementite (Fe_3C). Cooling rate significantly affects this ~~temperature~~ transformation. Fast cooling leads to carbide formation making it white cast iron. Slower cooling results in graphite precipitation.

which leads to gray cast iron. wedge shaped castings are ideal for studying this effect.

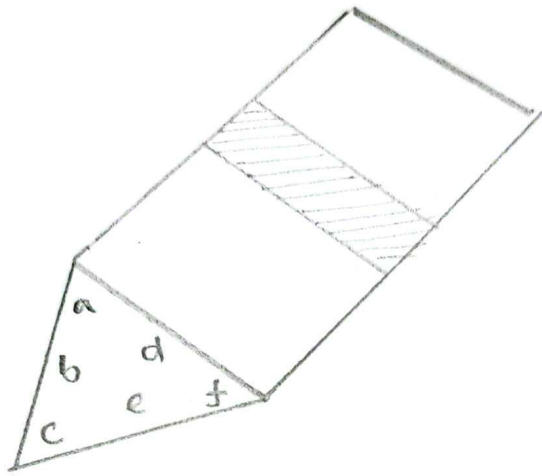


Fig: The wedge-shaped casting

At the corners of wedge-shaped casting, the cooling rate is higher than that of the centre of triangle. So, in the same casting, different types of cast iron is present with different hardness properties.

Inoculation, like ferrosilicon and ferromanganese plays a crucial role in controlling the microstructure.

Ferrosilicon influences graphitization, refines the structure and favors the formation of gray cast iron.

Ferromanganese acts as a deoxidizer

and helps control sulfur content.

It affects carbide formation and mechanical properties.

Additives such as deoxidizers and slag formers are introduced during melting to remove impurities.

Raw materials:

- ① Pig iron ($C = 3.8\%$, $Si = 2.5\%$, $Mn = 1.5\%$)
- ② Ferromanganese ($Mn = 80\%$)
- ③ Ferrosilicon ($Si = 60\%$)

Additives:

- ① Inoculant
- ② Deoxidizer
- ③ Slag former.

Experimental Procedure :

- ① Prepared green sand mould using the wedge-shaped pattern before the experiment.
- ② Melt about 10 kg Pig iron in a ladle using induction furnace. Removed ladle from the furnace after melting at about 1450°C . Added deoxidizers and slag formers to the ladle, mixed thoroughly and then skimmed the slag off. Measured the temperature of liquid before pouring it to the mould at 1300°C . Labeled as C-1.
- ③ Put the ladle back to the furnace with the remaining liquid. Removed ladle from the furnace after melting at about 1450°C . Added about 60g ferromanganese to the melt. Stir the melt thoroughly, remove the dross and pour at 1300°C . Labeled this as C-2.
- ④ Again same process but this time added ferrosilicon. casting named as C-3.
- ⑤ After cooling, removed castings from the mould. Polished for metallography and hardness measurement.

(vi) Took macro and micrograph and measured hardness.

Results:

(a) Hardness Test:

Sample - 03	Reading (HRB)	Avg. (HRB)
a	87.7	89
	90.5	
	88.8	
b	80.6	78.9
	81.4	
	74.7	
c	84.5	87.36
	87.6	
	90.0	